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SPOKEN LANGUAGES

Russian: Native  
English: B1  
Spanish: A1  
German: A1

CORE STRENGTHS

Platform ownership  
Production reliability  
Delivery automation  
Team leadership

ENVIRONMENTS

Cloud & bare metal  
Isolated, air-gapped  
Product & consulting  
Remote teams

DOMAIN EXPERIENCE

Finance & insurance  
Aviation & travel  
iGaming & anti-fraud

SENIOR PLATFORM & SRE ENGINEER

ILYA PAPOU

8 years in IT · 5+ in DevOps & SRE

Senior Platform & SRE Engineer with experience building and operating production Kubernetes platforms, automating software delivery, and leading DevOps teams. My background includes cloud and bare-metal infrastructure, security controls, observability, incident response, and disaster recovery. My current personal R&D focuses on agent infrastructure for software engineering, including MCP, A2A, and workflows with human review and approval.

PROFESSIONAL EXPERIENCE

RED ROSE TRAVELTECH — SENIOR SRE & PLATFORM ENGINEER

Aviation & travel

Mar 2025 — Sep 2026 · Contract · Remote

- Built and operated a unified bare-metal RKE2 Kubernetes platform for microservices, ML workloads and shared services. Used Terragrunt for infrastructure management.
- Established platform-wide Argo CD GitOps delivery with GitLab CI/CD and shared Helm charts for one-click deployment and configuration. Standardized ML service deployments and migrated them between platforms without downtime.
- Supported anti-fraud services. Customized Airflow Helm charts and extended DAG distribution mechanisms.
- Encrypted etcd, managed service accounts and centralized certificates with Vault. Prepared deployments to run without repository-stored credentials. Implemented mTLS, including Kafka and Redis client connections.
- Centralized packages and images in Nexus, with one Docker group endpoint and no direct external access. Added SAST, GitGuardian and Trivy checks to block vulnerable releases.
- Owned on-call incident response and documented recovery and preventive actions. Integrated kagent and Mattermost for AI-assisted engineering, managing agents and platform configuration through IaC and GitOps.

SBERBANK — HEAD OF DEVOPS

Finance & insurance

Jun 2023 — Mar 2025 · Full-time · Hybrid

- Built and led a DevOps team with four direct reports, coordinating an infrastructure group of ~10 people. Established onboarding, training, knowledge sharing, and regular technical reviews.
- Built cloud infrastructure from the ground up and provisioned ~8 Kubernetes clusters (20+ nodes total) with Terraform and GitOps. Improved availability through Multi-AZ and Multi-Region design, integrating cloud services with VMware infrastructure.
- Led service migration to the cloud with minimal downtime and controlled change management.
- Reduced infrastructure costs through automated shutdown of unused resources and usage governance.
- Built reusable Jenkins delivery pipelines for 30–40 microservices with Helm, security checks, tests, and approvals. Build and delivery took ~5 minutes for batches of more than six services.
- Introduced Vault, encrypted service communication, centralized monitoring, and tested disaster recovery procedures. Developed a Keycloak SSO proof of concept and coordinated requirements with security and development teams.

PUBLIC SECTOR — SENIOR DEVOPS ENGINEER

Public sector

Apr 2022 — Jun 2023 · Full-time · Remote

- Administered all development hypervisors and environments, including Hyper-V with 100+ virtual machines on Windows Server hosts in isolated enterprise and public-sector environments.
- Investigated production incidents involving Docker Swarm, RabbitMQ and PostgreSQL/Patroni. Maintained Airflow, resolved network issues and managed OpenVPN connectivity.
- Built Kubernetes on ALT Linux 9.2 in an air-gapped environment, wrote manifests and CI/CD pipelines, and automated offline releases.
- Administered Linux and automated operations with Ansible, Python and Bash, including Python scripts for Nexus administration.
- Maintained TeamCity, Nexus, Cassandra, Solr, ZooKeeper, HAProxy and ELK. Wrote deployment and configuration documentation for these services in isolated environments.

FLANT — SOFTWARE DEVELOPMENT & DEVOPS

Cross-industry IT consulting

Nov 2019 — Mar 2022 · Selected projects · Remote

- Configured, deployed and operated Java/Spring applications on Kubernetes. Developed reusable Helm charts with Go templates and contributed to Go development.
- Delivered AWS and Yandex Cloud infrastructure. Automated configuration with Ansible and used GitHub Actions to build and publish Docker images to Yandex Cloud.
- Managed production Prometheus, Grafana, ELK and PostgreSQL/Patroni. Built React, Next.js, Node.js and Express applications and improved website SEO.
- Created advertising materials, generated leads and managed a customer database for business automation equipment.

EARLIER EXPERIENCE

Information Security Intern · Telecommunications company · Feb 2018 — Nov 2019



AGENT ORCHESTRATION INFRASTRUCTURE

Jun 2026 – Present · Personal R&D · PoCs

- Personal R&D platform for developers and AI coding agents working on software and infrastructure tasks. Designed to preserve context across agent handoffs, connect tools and long-running workflows, and keep changes under human review and approval.
- Deployed platform components on Google Cloud and built four reusable Helm workload profiles with configuration validation and automated tests.
  - Built a Go Agent Host for native and Docker execution, with adapters for official Codex and Claude clients. Implemented A2A streaming, session continuation and cancellation, with TLS and short-lived access for host connections.
  - Added declarative agent configuration, model selection and separate development and production environments. Prepared Kubernetes workload isolation and default-deny network policies.
  - Designed A2A task handoffs and long-running workflows with kagent and Temporal, including retries and Human-in-the-loop approval gates for code and infrastructure changes.
  - Implemented MCP tool allowlists and a Go policy API with strict request validation, deny-by-default decisions and audit records. Pinned runtime skills to immutable artifact digests.
  - Built and ran container-image and Helm-chart release workflows. Prepared Google Cloud scanning and approval-based promotion pipelines.
  - Tested release policies and validated Agent Host contracts with simulated Codex and Claude providers.

Code [Platform](#) [Agent runtime](#) [kagent fork](#) [kagent integration](#) [Mattermost fork](#) [Image build](#)

TECHNICAL SKILLS

Platform & delivery

Linux · Kubernetes · RKE2 · Docker  
Terraform · HCP Terraform Stacks  
Helm · Argo CD · Ansible  
GitLab CI/CD · Jenkins  
GitHub Actions · Nexus

Cloud infrastructure

AWS · GCP  
S3 · Lambda · EKS · IAM  
GKE · Cloud SQL · GCS  
Artifact Registry · Cloud Build & Deploy  
IAM · WIF · Secret Manager  
Cloud KMS · Pub/Sub · Cloudflare

Software engineering

Go · Python · Bash · JavaScript  
React · Next.js · Node.js · Express  
Java · Spring · C++  
OpenAPI · gRPC  
Schema validation · contract tests

Security & reliability

Vault · ESO · Keycloak · mTLS · Kyverno  
GitGuardian · Trivy  
Prometheus · VictoriaMetrics  
Grafana · ELK · OpenTelemetry  
Tailscale · OPNsense

ML & data infrastructure

Airflow · Slurm · JupyterHub · MLflow  
PostgreSQL · Patroni · Redis  
Kafka · RabbitMQ · Cassandra  
MinIO · VectorDB · BigQuery

Agent infrastructure & R&D

kagent · agentgateway  
MCP · A2A · RAG  
Temporal · Human-in-the-loop  
LangChain · ADK  
Codex · Claude Code · Gemini  
vLLM · Langfuse

HOME AEROPONICS & IOT AUTOMATION

Aug 2024 – Jan 2025 · Personal R&D

- Designed and built an integrated aeroponic system: drew wiring schematics, soldered electronics, and connected ESP32, Arduino, Raspberry Pi, sensors and actuators.
- Developed C++ sensor firmware and Python automation for climate control, mist generation, water supply and circulation, tank-to-tank transfers and leak detection.
- Automated lighting and collected light-quality data to guide adjustments. Used computer vision to assess experimental results through changes in leaves.
- Connected ESP and Zigbee devices through Mosquitto and Zigbee2MQTT using JSON over MQTT, with Home Assistant as the control, monitoring and alerting interface.
- Automated Raspberry Pi setup with Ansible, prototyped pH, TDS and EC monitoring, and began preparing Prometheus metrics collection.

Code [Controllers](#) [Sensor firmware](#)

ZERO-TRUST MESH & OPEN-SOURCE NGFW

Jan 2025 – Mar 2025 · Personal R&D

- Built an encrypted, multi-region Tailscale lab mesh linking devices and private networks across cloud, bare metal and homelabs.
- Automated network configuration and enabled one-click onboarding of new nodes.
- Managed all mesh ACLs through GitOps, with distinct user and service-account permissions for DNS discovery and service access.
- Tested OPNsense as an NGFW for subnet routing, network segmentation and traffic filtering.

Code [Ansible](#) [Access policy](#) [GCP network lab](#)